

100 DAYS OF TQM : CASE STUDY DAY 29/100

Samsung's Total Quality Transformation & Management Strategy

A Strategic Analysis of Executive Leadership, Worker Empowerment, Taguchi Robust Engineering, and International Quality Standards

1. STRATEGIC CONTEXT & LEADERSHIP TRANSFORMATION

In 1993, Samsung executed one of the most remarkable strategic turnarounds in modern industrial history under Chairman Lee Kun-hee. Shifting the core operational paradigm from volume-driven production to absolute quality supremacy, Chairman Lee declared at the historic Frankfurt Declaration: *"Change everything except your wife and children."* This wasn't merely a slogan, but a top-down executive directive recognizing that accepting even a 1% defect rate fundamentally compromises 100% of customer trust.

Strategic Alignment with Deming's 14 Points:

- 1. Adopt the New Philosophy:** Management established that change is an absolute operational law. Accepting the new definition of quality required a complete cultural mindset shift across all management layers.
- 2. Eliminate Slogans & Exhortations:** Transitioning from passive quality goals to active structural reform, enforcing zero tolerance for compromised standards.

2. SHOP-FLOOR EMPOWERMENT & COST OF QUALITY (COQ)

- Worker Empowerment & Andon Line-Stop Authority (Deming's "Drive Out Fear"):** Senior executives cannot constantly monitor shop-floor operations. Empowering assembly line operators with complete authority to trigger line stops (Andon system) upon detecting non-conformance eliminates fear, encouraging workers to expose and rectify defects at the source rather than hiding them due to production pressure.
- Risk Management & Failure Cost Trade-off:** Allowing non-conforming items to progress down the assembly line incurs escalating costs. Catching defects on the shop floor causes *Internal Failure Cost*, but letting defective products reach consumers triggers catastrophic *External Failure Cost*—destroying warranty budgets, brand reputation, customer trust, and long-term retention.
- Zero Defects & Quality Sacrifice (Crosby's Philosophy):** In 1995, following defect reports in the SH-770 mobile phone, Chairman Lee ordered the public burning of 150,000 defective devices valued at \$50 million in front of 2,000 employees. Aligning with Philip Crosby's *Zero Defects* (*"Do it right the first time"*), this extreme decision demonstrated that short-term financial sacrifice is mandatory to protect long-term brand equity and customer faith.

3. ROBUST ENGINEERING & PRODUCTIVITY ENHANCEMENT

- Taguchi's Robust Design & Design of Experiments (DOE):** Quality must be designed into the product from inception. By utilizing Taguchi's Design of Experiments (DOE), Samsung optimizes controllable factors to make product performance immune to uncontrollable "noise" factors (e.g., environmental heat, user handling). This directly reduces non-conformance, lowers failure costs, minimizes breakdown maintenance, and secures a resilient zero-defect architecture.
- Productivity & Root Cause Problem Solving:** To maximize line productivity without compromising safety, structured problem-solving methodologies are deployed:
 - 5-Why Analysis:** Iteratively drilling down to uncover the true root cause behind line stoppages.
 - Fishbone (Ishikawa) Diagram:** Categorizing potential causes across Man, Machine, Material, Method, Measurement, and Environment.
 - Pareto Analysis (80/20 Rule):** Isolating the "vital few" defects that cause 80% of production downtime.

4. QUALITY GOVERNANCE & INTERNATIONAL STANDARDS (ISO)

- **ISO 9001:2015 (Quality Management Systems - Requirements):** Provides the foundational framework for risk-based thinking, process approach, customer focus, and standardized operational procedures across global manufacturing facilities.
- **ISO 9004:2018 (Quality Management - Quality of an Organization - Guidance to Achieve Sustained Success):** Extends beyond basic ISO 9001 compliance. It provides management guidance for continuous self-assessment, organizational maturity, and long-term sustained success in a rapidly changing business environment.
- **ISO 19011:2018 Guidelines for Auditing Quality Systems (Three-Tier Audit Governance):**
 - **1st Party Audit (Internal Audit):** Regular shop-floor audits conducted internally to verify daily process adherence and line compliance.
 - **2nd Party Audit (Supplier Audit):** Rigorous audits performed on component suppliers and vendors to ensure raw material quality at the source.
 - **3rd Party Audit (External Certification Audit):** Independent audits conducted by accredited certification bodies to validate full ISO compliance and global benchmark standards.

5. APPLIED TQM CONCEPTS, DEFINITIONS & OPERATIONAL MAPPING

TQM CONCEPT / TOOL	SHORT ENGINEERING / MANAGEMENT DEFINITION	SAMSUNG OPERATIONAL MAPPING
Frankfurt Declaration (1993)	Top-down executive directive prioritizing quality supremacy over production volume.	Chairman Lee's "Change Everything" mandate shifting organizational culture.
Deming's Philosophy	14-point management principles focusing on systemic change, driving out fear, and adopting new quality definitions.	Worker empowerment via Andon line-stop system and elimination of meaningless slogans.
Crosby's Zero Defects	Quality philosophy stating "Do it right the first time" and recognizing quality is free compared to cost of defects.	1995 phone burning incident destroying \$50M non-conforming inventory to enforce zero tolerance.
Cost of Quality (CoQ)	Financial metric balancing Prevention/Appraisal costs against Internal and External Failure costs.	Absorbing internal failure costs via line stops to prevent devastating external warranty and trust losses.
Taguchi Robust Design & DOE	Statistical method optimizing controllable factors during design to make products insensitive to noise factors.	Front-end smartphone engineering reducing defect variation and breakdown maintenance.
5-Why & Fishbone Diagram	Root cause analysis tools systematically identifying underlying drivers of defects across 6M categories.	Assembly line defect troubleshooting to enhance line productivity and operational uptime.
Pareto Analysis (80/20)	Statistical decision tool prioritizing the 20% critical root causes responsible for 80% of quality issues.	Focusing engineering resources on vital defect categories causing high yield loss.
ISO 9001:2015	Standard specifying requirements for a formal QMS based on risk management and PDCA cycles.	Baseline quality certification across all global assembly plants and SMT lines.
ISO 9004:2018	Management standard providing guidance for organizational maturity and sustained long-term success.	Strategic self-assessment framework for continuous organizational maturity and growth.
ISO 19011 Three-Tier Audit	Comprehensive audit guideline covering 1st Party (Internal), 2nd Party (Supplier), and 3rd Party (External) audits.	Strict multi-layer quality governance from component suppliers to finished goods shipping.

Disclaimer: This 2-page case study is developed strictly for educational purposes as part of the 100 Days Of TQM learning challenge. Author: Jagriti Singh, Mechanical Engineering / Quality Engineer. This analysis evaluates management, TQM, and ISO standards from an academic and industrial engineering perspective.